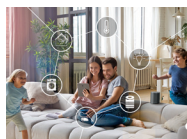


aspect ratios for touch displays in cars. This new offering includes additional functional safety support requested by OEMs. The MXT1296M1T can reconfigure its driving and receiving touch channels to match the exact screen format, from 1:1 to 5:1 aspect ratio, including the popular 8:3 automotive aspect ratio. This feature enables the customer to efficiently use the number of touch channels available, without the need to select a larger, more expensive touch controller.

Microchip Technology
www.microchip.com

Bluetooth LE with SoC

The new AIROC Bluetooth LE and 802.15.4 family help companies quickly bring low-power, high-performing Matter products to market. The AIROC CYW30739 Bluetooth LE and 802.15.4



system on chip (SoC) is a reliable, secure, and scalable solution to connect low-power devices in smart homes. The powerful combination of complementary Bluetooth LE and 802.15.4 protocols enhances the performance of smart home products with seamless interoperability, while enabling end-to-end encrypted communication between individual devices in a Matter network.

Infineon Technologies
www.infineon.com

SoC for 5G radio

The RadioVerse System-on-Chip (SoC) series provides radio unit (RU) developers with an agile and cost-effective platform to create the most energy-efficient 5G RUs in the industry. The new SoC series provides advanced RF signal processing with expanded digital functionality and RF capacity that greatly improves 5G RU performance and energy



efficiency. The ADRV9040 is the first in the new RadioVerse SoC series. It offers eight transmit and receive channels of 400MHz bandwidth and integrates advanced digital signal processing, including carrier digital up-converters, carrier digital down-converters, crest factor reduction, and digital pre-distortion.

Analog Devices
www.analog.com

NTC thermistor chip

The new L860 NTC thermistor chip can be directly embedded into power modules and used within sintering



and AI-heavy wire bonding. Its characteristics correspond to the common MELF-R/T curve with R100 of 493Ω. The lead-free chip is designed for a wide temperature range of -55° to +175°C and has dimensions of 1.6 × 1.6 × 0.5mm. Unlike conventional SMD NTC chips, the leadless NTC chip is mounted horizontally. For electrical bonding, the product is fitted on the bottom with a Ni/Ag thin-film electrode for sintering on the DCB (direct copper bonded) board of power modules, while the upper electrode comprises a Ni/Au thin-film electrode for aluminium wire bonding.

TDK Corporation
www.tdk.com

8K 120Hz TV chip

The new Pentonic 2000 powers the next-generation flagship 8K TVs. It is manufactured using TSMC's N7 (7nm-class) process, delivering unparalleled performance and power efficiency. This all-in-one chip supports 8K 120Hz displays and can even support 144Hz for gaming PC hardware and the ninth generation of consoles. Pentonic 2000 is the first chip with an integrated 8K 120Hz MEMC engine. The chip's built-in high-performance MediaTek APU (AI processor) powers MediaTek's new 8K AI Super Resolution technology that intelligently upscales lower-reso-

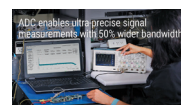


lution content to the display's native resolution, while also performing real-time image quality enhancements.

MediaTek
www.mediatek.com

Tiny 24-bit ADC

The ADS127L11 is the smallest 24-bit wideband analogue-to-digital converter (ADC) that delivers industry-leading signal-measurement precision at wider bandwidths than competing



ADCs. It achieves ultra-precise data acquisition in a 50% smaller package, significantly optimising power consumption, resolution, and measurement bandwidth for a wide range of industrial systems. Designers can use wideband mode to improve AC measurement resolution with 50% wider bandwidth, data rates as high as 400 kSPS, and a 30% higher signal-to-noise ratio than competing data converters, while minimising noise at high frequencies.

Texas Instruments
www.ti.com

MODULES & BOARDS

LTE vehicle tracker

The new AK300 LTE vehicle tracker helps provide real-time positioning, related driving safety management and other monitoring information for higher-value telematics applications.



It is enabled to support applications including tire pressure and temperature monitoring via Bluetooth, vehicle information retrieval for real-time position, fuel consumption, mileage and vehicle status anytime, and driver identification using RFID reader and tag. Additional applications include SOS emergency calls and photography.

ATrack Technology
www.atrak.com.tw

DC-DC converters

The BMR4920100/001 is rated at